

SELF-ATTENTIVE VAD: CONTEXT-AWARE DETECTION OF VOICE FROM NOISE

Yong Rae Jo^{*1}, Young Ki Moon^{*1,2}, Won Ik Cho³, Geun Sik Jo²

¹Voithru Inc., ²Inha University, ³Seoul National University

ABSTRACT

Recent voice activity detection (VAD) schemes have aimed at leveraging the decent neural architectures, but few were successful with applying the attention network due to its high reliance on the encoder-decoder framework. This has often let the built systems have a high dependency on the recurrent neural networks, which are costly and sometimes less context-sensitive considering the scale and property of acoustic frames. To cope with this issue with the self-attention mechanism and achieve a simple, powerful, and environment-robust VAD, we first adopt the self-attention architecture in building up the modules for voice detection and boosted prediction. Our model surpasses the previous neural architectures in view of low signal-to-ratio and noisy real-world scenarios, at the same time displaying the robustness regarding the noise types. We make the test labels on movie data publicly available for the fair competition and future progress.

Index Terms— voice activity detection, self-attention, real-world noise

1. INTRODUCTION

Voice activity detection (VAD) is a simple task of detecting speech in an audio input but has an essential role in real-world audio processing. For instance, by using VAD, noise and silence can be eliminated when speech recognition is performed for a noisy audio [1], and in speaker verification, one can prevent misunderstanding of noise as a speaker [2]. Traditional VAD, mainly concerns the performance at a low signal-to-noise ratio (SNR), but the recent trend pays attention to performance in messy real-world audio[3, 4].

The early VAD research scrutinized acoustic features such as zero-crossing rate, pitch detection, and energy entropy [5]. In these methods, the heuristic setting of threshold made it challenging to establish a standard that works well in real-world data. In the advanced statistical methods [6, 7], the presence of speech was predicted by assuming the distribution of non-speech and speech based on the history of each, e.g., via the Hidden Markov model. Though they have shown the notable performance compared to the early approaches, due to Markovian assumption, the probability regarding the less

frequent sequence pattern can be determined low, accordingly hindering the generalization for the unseen data.

Recently, deep learning (DL)-based approaches have shown remarkable performance in various downstream tasks. It enables us to handle large-scale data due to its scalability and displays better generalization performance than the previous machine learning methods. VAD has also been boosted by this trend, with mixed acoustic features as an input of a neural network [8, 9], or by modeling contextual information with prevalent neural architectures such as deep neural network (DNN) [10], convolutional neural network (CNN) [11, 12], and recurrent neural network (RNN) [3, 13].

Techniques for extracting meaningful features within context using CNN [11, 12] or boosted DNN[10] are known to have a decent performance. However, modeling contextual information using RNNs has not been successful. It was known difficult to capture a global audio context even using bidirectional architecture, which hinders RNN-based methods from outpacing other neural architectures in detecting speech with sequential features. In this regard, ACAM [14] proposed a VAD using LSTM and attention mechanism [15], but the attention mechanism, primarily used to model the correlation between the hidden vectors regarding the encoder and decoder [16], seemed less suitable for the VAD task which aims at frame-wise classification.

Nonetheless, since the attention mechanism has exhibited practically high performance in many circumstances that involve the context, we inferred that we could leverage such properties as well in VAD. We hypothesized that modeling audio context between input acoustic frames via self-attention [17] would be effective for understanding the audio as a token sequence with noise and voice. Through the experiment, we check that the proposed method outperforms the previous neural network and attention-based approaches.

The contribution of this study is as follows:

- We propose the modeling of contextual information between input acoustic frames, for the first time using the self-attention mechanism.
- We demonstrate our approach has an advantage over other models in various SNR scenarios and real-world audio.
- We publicly release the movie sound dataset¹ tagged

^{*}Both authors contributed equally to this manuscript.

¹To be disclosed (after acceptance)

with voice activity timestamp, for a fair evaluation and future research progress.

2. PROPOSED MODEL

The self-attentive VAD proposed in this paper takes an acoustic feature vector extracted from the audio as an input, and decides whether each frame contains speech (1) or not (0). The architecture is depicted in Fig. 1.

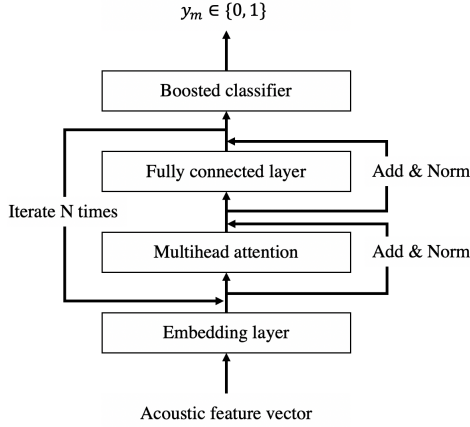


Fig. 1. Self-attentive VAD

2.1. Input Feature

The input acoustic frame is obtained by hopping an audio window of length 25ms with 10ms sliding. The frame is converted to acoustic feature $x_m (m = 1, \dots, M)$ through the feature extraction process, where m is the input time index. $y_m \in \{0, 1\}$ denotes the label of each acoustic feature.

For efficient modeling of contextual information between the input frames, the features around x_m is augmented to make up a new acoustic feature vector v_{x_m} as in (1). The same holds for v_{y_m} as in (2), where w and u are the hyper-parameters.

$$v_{x_m} = [x_{m-w}, x_{m-(w-u)}, \dots, x_m, \dots, x_{m+(w-u)}, x_{m+w}] \quad (1)$$

$$v_{y_m} = [y_{m-w}, y_{m-(w-u)}, \dots, y_m, \dots, y_{m+(w-u)}, y_{m+w}] \quad (2)$$

2.2. Embedding Layer

v_{x_m} is converted into an embedding vector h_{x_m} to be fed as an input of the self-attention network. In this process, the feature vector passes the affine layer and is activated by ReLU, consequently going through sinusoidal positional encoding to add ordinal information to embedding.

$$h_{x_m} = \text{sinusoidal}(\max(0, \text{Affine}(v_{x_m}))) \quad (3)$$

2.3. Self-Attention Network

The self-attention network is composed of i) multi-head self-attention, ii) a sublayer that performs residual connection and normalization, and iii) a point-wise feed-forward layer.

Multi-head attention uses multiple attention networks to prevent bias of the singleton case. Extended from the single self-attention that receives the embedding vector h_{x_m} and calculates the score between h_{x_m} , the multi-head attention $W_i^{h_{x_m}}$ adopts matrix multiplication with h_{x_m} .

$$\text{Attention}(h_{x_m}) = \left(\frac{h_{x_m} h_{x_m}^T}{\sqrt{d_{n_{x_m}}}} \right) h_{x_m} \quad (4)$$

$$\text{Multihead}(h_{x_m}) = [\text{head}_1, \dots, \text{head}_n] W^0 \quad (5)$$

$$\text{where head}_i = \text{Attention}(h_{x_m} W_i^{h_{x_m}})$$

Sublayer performs residual connection and layer normalization to mitigate overfitting and smoothe the gradient.

Point-wise feed-forward layer executes two affine operations and ReLU activation. In general, the input hidden vector dimension is increased in the first affine operation, and then reduced to the original size consequently. That is, the layer prevents the internal bias of the model while controlling the dimension of the hidden vectors.

$$\text{FFL}(x) = \max(0, W_1 x + b_1) W_2 + b_2 \quad (6)$$

2.4. Boosted Classifier

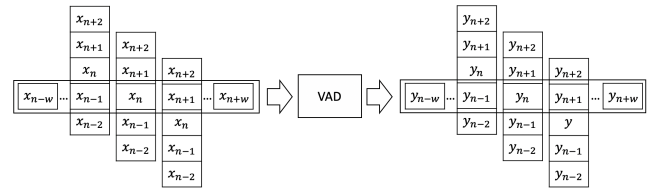


Fig. 2. Boosted classifier

Boosted classifier, a sort of ensemble, determines the index of each frame using average, not just prediction of single input acoustic vector. The detailed operation is as below:

$$\hat{y}_n = \frac{y_n^{(-w)} + \dots + y_n^{(-1)} + y_n^{(0)} + y_n^{(1)} + y_n^{(w)}}{2w + 1} \quad (7)$$

3. EXPERIMENTS

3.1. Dataset

To verify the generalizability of the proposed model, we constructed training and test data with different sources. For training data, TIMIT [18] was used with five background

	# parameter	MUSAN	MIX
bDNN	1.6M	0.9764	0.9798
ACAM	870K	0.9730	0.9744
SA	370K	0.9760	0.9812

Table 1. Model parameter and validation results

noises: MUSAN [19], MR, Sound effect library², Youtube-SoundEffect³, and YoutubeEditorSoundEffect. At this time, .wav MR files with the keyword “instrumental” in the title were crawled from online, and the samples adequate as background were chosen via human listening. For TIMIT, two seconds of silence was added before/after each file.

Two scenarios were created from the above data. The first scenario (MUSAN) was generated by adding noise data in MUSAN to TIMIT voice, and the second scenario (MIX) was created by random sampling from the other four datasets. Sampled noise segments were placed to overlap with each other. In both noise scenarios, the silence was placed periodically, by 10% length of the inserted noise. While injecting noise, SNR was randomly selected between -6 and 12 in both scenarios. 5% of the training data was used for validation.

For test, four datasets were prepared: Noisex-TIMIT which combines Noisex-92 [20] with TIMIT, ENVIRONMENT (park, bus stop, construction site, and room) [14], MOVIE [21], and YOUTUBE [22].

In the case of Noisex-92 test data, SNR {-10,-5,0,5,10} was set and aggregated with TIMIT data in the same manner as training data. ENVIRONMENT, provided by [14], includes the recordings at the four locations, done with a mobile device. MOVIE was selected from the list used in [21], guaranteeing the balance between the movie genres, and the speech/non-speech label was created by referring to the .srt file. For YOUTUBE, videos were randomly chosen from YOUTUBE-8M [22] to fulfill a total of 24 hours in scale, accompanied by manual annotation.

3.2. Baseline Model

To compare the performance of the proposed model, bDNN [10] and ACAM [14] were used as the baseline, based on the borrowed hyperparameters from the original paper. AUC was used for the evaluation, and the test was conducted using the checkpoint having the highest AUC in the validation dataset after 30 epochs of the training.

3.3. Settings

For self-attentive VAD, multi-resolution cochleagram (MRCG) [23] was used for the acoustic feature extraction, with the embedding dimension 128. The attention layer width was set to 64 and the point-wise feed-forward layer to 256. The batch size was set to 4,096 and the learning rate to 1e-4, with the

²<https://www.sound-ideas.com/Product/51/General-Series-6000-Combo>

³<https://www.youtube.com/audiolibrary/soundeffects>

Noise Scenario		MUSAN			MIX		
Noise	SNR	bDNN	ACAM	SA	bDNN	ACAM	SA
babble	-10	0.7092	0.7804	0.7768	0.8028	0.7983	0.8150
	-5	0.9064	0.9023	0.8904	0.9286	0.9146	0.9298
	0	0.9609	0.9613	0.9529	0.9791	0.9689	0.9786
	5	0.9814	0.9819	0.9769	0.9917	0.9863	0.9911
	10	0.9900	0.9905	0.9889	0.9955	0.9928	0.9949
f16	-10	0.9224	0.9093	0.9234	0.8806	0.9275	0.9369
	-5	0.974	0.9709	0.9763	0.9627	0.9736	0.9810
	0	0.9884	0.9877	0.9898	0.9878	0.9876	0.9911
	5	0.9918	0.9922	0.9926	0.993	0.992	0.9938
	10	0.9948	0.994	0.9949	0.9944	0.9945	0.9952
factory1	-10	0.8842	0.8601	0.8901	0.9241	0.8852	0.9235
	-5	0.9564	0.9451	0.9582	0.9727	0.9557	0.9722
	0	0.9813	0.9777	0.9821	0.9872	0.9806	0.9876
	5	0.9892	0.9884	0.9894	0.9914	0.9887	0.9917
	10	0.9918	0.9914	0.9924	0.9932	0.9919	0.9934
factory2	-10	0.979	0.9612	0.9807	0.9795	0.9642	0.9806
	-5	0.988	0.9849	0.9887	0.9902	0.985	0.9909
	0	0.9929	0.9924	0.9935	0.9941	0.9918	0.9944
	5	0.9952	0.995	0.9956	0.996	0.9948	0.9962
	10	0.9966	0.9964	0.9968	0.9973	0.9966	0.9974
leopard	-10	0.9937	0.9922	0.9935	0.9948	0.9935	0.9956
	-5	0.9959	0.9954	0.9961	0.9966	0.9962	0.9975
	0	0.9975	0.9968	0.9975	0.9977	0.9975	0.9982
	5	0.9984	0.9975	0.9985	0.9988	0.9983	0.9987
	10	0.9989	0.9981	0.9990	0.9995	0.9987	0.9992
pink	-10	0.9623	0.9394	0.9670	0.9689	0.9584	0.9731
	-5	0.9824	0.9785	0.9862	0.988	0.9825	0.9893
	0	0.9888	0.9887	0.9914	0.9918	0.9898	0.9933
	5	0.9919	0.9919	0.9933	0.9935	0.9924	0.9944
	10	0.9935	0.9928	0.9941	0.9944	0.9942	0.9952

Table 2. Test result - Noisex92

dropout rate 0.3 and the weight decay 1e-5. The gradient clipping was set between -1 and 1.

4. RESULTS AND ANALYSIS

4.1. Model Parameters and Validation

The parameter of the proposed model is 370K which is much smaller than ACAM’s 870K and bDNN’s 1.6M, with similar or better performance (Table 1). Though bDNN showed good validation performance in MUS that has relatively little noise, the proposed model outweighs in much noisy MIX scenario. This suggests that the proposed model is robust enough to maintain the performance in a harsh environment while lessening the parameter size.

4.2. Conventional Noise Scenarios

Detailed on the MUSAN and MIX results in conventional Noisex92 scenarios, the proposed model showed decent performance except for *babble* (Table 2). The excellence was most significant in *factory1* and *factory2*, again proving the strength of the proposed model against loud noise. A similar result for comparably quiet *pink* scenario displays the robustness of the proposed model to conventional noise scenarios.

The exception was in *babble*, where ACAM outperforms in MUSAN unlike other cases. It seems to originate in that the adaptive contextual attention of ACAM, which consistently refers to input data, might have enhanced the detec-

dataset	Noise scenarios	MUSAN			MIX		
		bDNN	ACAM	SA	bDNN	ACAM	SA
ENVIRONMENT	park	0.9861	0.9849	0.9880	0.9818	0.9820	0.9850
	bus stop	0.9830	0.9820	0.9833	0.9811	0.9732	0.9827
	construction site	0.9966	0.9948	0.9968	0.9956	0.9946	0.9967
	room	0.9906	0.9902	0.9920	0.9901	0.9891	0.9903
MOVIE	Armageddon	0.8785	0.8464	0.8793	0.8696	0.8380	0.8755
	Dead poets society	0.863	0.8566	0.8647	0.8591	0.8564	0.8628
	Forrest gump	0.8719	0.8648	0.8706	0.8691	0.8643	0.8734
	Independence day	0.8245	0.8073	0.8289	0.8214	0.7905	0.8250
	Legally blonde	0.8181	0.8192	0.8164	0.8261	0.8214	0.8226
	Saving private ryan	0.8465	0.8238	0.8523	0.8520	0.8250	0.8579
YOUTUBE	-	0.8616	0.8464	0.8702	0.8546	0.8584	0.8715

Table 3. Test result - Real-world audio dataset

tion of voice from other voice-like (babble) noises. However, such characteristics failed to bring advantage in the other scenario. In the MIX scenario, the proposed model and bDNN surpasses ACAM for *babble*, which shows that the performance of ACAM is prone to be sensitive to the background noise. Also, we claim that the proposed is more effective in harsh conditions (low SNR), which is to be desired in real-world applications.

4.3. Real-World Audio

In ENVIRONMENT, the performance of the model in all locations is significant, considering that the test incorporates the type of noise that is absent in the train set (Table 3). Besides, for MOVIE, except for two movies (*Forrest Gump* and *Legally Blonde*), the proposed model exceeds the others and especially ACAM, particularly for *Armageddon*, *Independence day*, and *Saving private ryan*. The proposed structure seems to guarantee the wide-range generalizability over the others, provided that the movie data incorporates considerably more amount of environmental audio events than the conventional noises. We assume that our model leverages the context-aware representation of self-attention that have benefited token-level encoding, which here accommodates the room for comprehending the ‘audio words’ which are connoted in the acoustic frames in the shape of the signal.

In YOUTUBE, the most far from the canonical noise scenarios, the proposed model performed noticeably better than the others. Since Youtube consists of videos from world-wide users, capturing voice activity within there means high generalizability regarding real-world audio processing. This verifies the utility of our approach, which is directly applicable to tasks such as automatic captioning of videos, movie subtitling, or outdoor call quality enhancement.

4.4. Summary

Overall, we confirmed that our model surpasses other existing models both in Noisex92 and the real-world scenarios, better with MIX than MUSAN. Our model achieves decent

performance for the challenging test set, by discerning ‘what should the voice be like in specific audio environment’ using self-attentive training and various noise.

Let us turn to the genuine meaning of ‘self-attentive’ and ‘context’ in recognition of voice presence. Voice, in the experiments, differs from the noises that come from the human crowd (*babble*), machines (*factory*, *f16*, *leopard*), signals (*pink*), and life environments (*park*, *bus stop*, *construction*, *room*). Given each ‘context’, namely the bag of audio events, how can a machine be aware of some additive mixed voices?

Here is our explanation: just as the subwords making up a local semantics, an audio semantics is constituted by the frames with relevant signal information for the acoustic events. As in language where the long-term dependency is modeled to determine the influence of a key to a certain query, self-attention on a frame sequence tells if some events can be beneficial to the voice prediction of a certain frame. It seems to hold, and is expected to be more meaningful when the audio semantics is engaged with a complicated local window.

5. CONCLUSION

In this paper, we proposed a self-attentive VAD that models contextual information between acoustic frames to benefit the prediction. The training went through two scenarios with different noise levels, with the evaluation done on conventional and real-world datasets. The proposed model outperforms the baselines with fewer parameters, showing robustness for the harsh, real-world scenarios. The test annotation on movie files is to be distributed for fair comparison and future progress.

6. REFERENCES

- [1] Javier Ramírez, José C Segura, Carmen Benítez, Angel De la Torre, and Antonio Rubio, “An effective subband osf-based vad with noise reduction for robust speech recognition,” *IEEE Transactions on Speech and Audio Processing*, vol. 13, no. 6, pp. 1119–1129, 2005.

- [2] T. Kinnunen and P. Rajan, "A practical, self-adaptive voice activity detector for speaker verification with noisy telephone and microphone data," in *2013 IEEE International Conference on Acoustics, Speech and Signal Processing*, 2013, pp. 7229–7233.
- [3] Florian Eyben, Felix Weninger, Stefano Squartini, and Björn Schuller, "Real-life voice activity detection with lstm recurrent neural networks and an application to hollywood movies," in *2013 IEEE International Conference on Acoustics, Speech and Signal Processing*. IEEE, 2013, pp. 483–487.
- [4] Neville Ryant, Mark Liberman, and Jiahong Yuan, "Speech activity detection on Youtube using deep neural networks,," in *INTERSPEECH*. Lyon, France, 2013, pp. 728–731.
- [5] Philippe Renevey and Andrzej Drygajlo, "Entropy based voice activity detection in very noisy conditions," in *Seventh European Conference on Speech Communication and Technology*, 2001.
- [6] Jongseo Sohn, Nam Soo Kim, and Wonyong Sung, "A statistical model-based voice activity detection," *IEEE Signal Processing Letters*, vol. 6, no. 1, pp. 1–3, 1999.
- [7] Joon-Hyuk Chang, Nam Soo Kim, and Sanjit K Mitra, "Voice activity detection based on multiple statistical models," *IEEE Transactions on Signal Processing*, vol. 54, no. 6, pp. 1965–1976, 2006.
- [8] Thomas Drugman, Yannis Stylianou, Yusuke Kida, and Masami Akamine, "Voice activity detection: Merging source and filter-based information," *IEEE Signal Processing Letters*, vol. 23, no. 2, pp. 252–256, 2015.
- [9] Stefan Meier and Walter Kellermann, "Artificial neural network-based feature combination for spatial voice activity detection,," in *INTERSPEECH*, 2016, pp. 2987–2991.
- [10] Xiao-Lei Zhang and DeLiang Wang, "Boosting contextual information for deep neural network based voice activity detection," *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 24, no. 2, pp. 252–264, 2015.
- [11] Ruben Zazo, Tara N Sainath, Gabor Simko, and Carolina Parada, "Feature learning with raw-waveform cldnns for voice activity detection," in *INTERSPEECH*, 2016, pp. 3668–3672.
- [12] Yasunari Obuchi, "Framewise speech-nonspeech classification by neural networks for voice activity detection with statistical noise suppression," in *2016 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2016, pp. 5715–5719.
- [13] Thad Hughes and Keir Mierle, "Recurrent neural networks for voice activity detection," in *2013 IEEE International Conference on Acoustics, Speech and Signal Processing*. IEEE, 2013, pp. 7378–7382.
- [14] Juntae Kim and Minsoo Hahn, "Voice activity detection using an adaptive context attention model," *IEEE Signal Processing Letters*, vol. 25, no. 8, pp. 1181–1185, 2018.
- [15] Dzmitry Bahdanau, Kyunghyun Cho, and Yoshua Bengio, "Neural machine translation by jointly learning to align and translate," in *3rd International Conference on Learning Representations, ICLR 2015*, 2015.
- [16] Ilya Sutskever, Oriol Vinyals, and Quoc V Le, "Sequence to sequence learning with neural networks," in *Advances in Neural Information Processing Systems*, 2014, pp. 3104–3112.
- [17] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Lukasz Kaiser, and Illia Polosukhin, "Attention is all you need," in *Advances in Neural Information Processing Systems*, 2017, pp. 5998–6008.
- [18] J. Garofolo, Lori Lamel, W. Fisher, Jonathan Fiscus, D. Pallett, N. Dahlgren, and V. Zue, "TIMIT acoustic-phonetic continuous speech corpus," *Linguistic Data Consortium*, 11 1992.
- [19] David Snyder, Guoguo Chen, and Daniel Povey, "MUSAN: A music, speech, and noise corpus," *arXiv preprint arXiv:1510.08484*, 2015.
- [20] Andrew Varga and Herman JM Steeneken, "Assessment for automatic speech recognition: II. NOISEX-92: A database and an experiment to study the effect of additive noise on speech recognition systems," *Speech communication*, vol. 12, no. 3, pp. 247–251, 1993.
- [21] Mats Sjöberg, Bogdan Ionescu, Yu-Gang Jiang, Vu Lam Quang, Markus Schedl, and Claire-Hélène Demarty, "The mediaeval 2014 affect task: Violent scenes detection," Oct. 2014.
- [22] Sami Abu-El-Haija, Nisarg Kothari, Joonseok Lee, Paul Natsev, George Toderici, Balakrishnan Varadarajan, and Sudheendra Vijayanarasimhan, "Youtube-8M: A large-scale video classification benchmark," *arXiv preprint arXiv:1609.08675*, 2016.
- [23] Jitong Chen, Yuxuan Wang, and DeLiang Wang, "A feature study for classification-based speech separation at low signal-to-noise ratios," *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 22, no. 12, pp. 1993–2002, 2014.